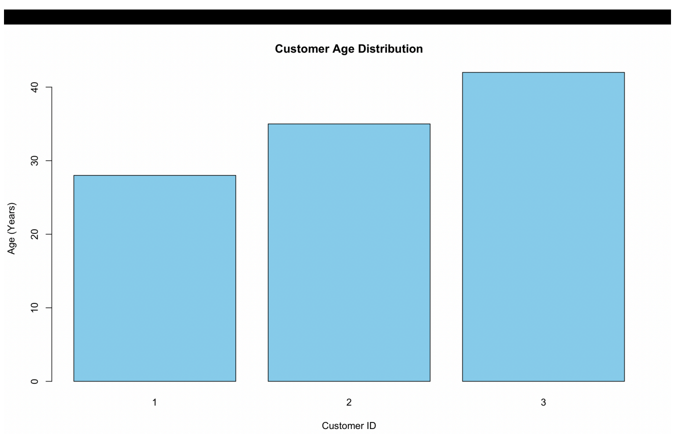
LAB EXPERIMENTS
Date: 29.07.2026
Questions : 16 - 20

Set 16: Customer Demographics Analysis
Q1. Bar Chart – Customer Ages

Program:

```
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(
    28,
    35,
    42
  ),
  Gender = c(
    "Female",
    "Male",
    "Female"
  ),
  Income = c(
    50000,
    60000,
    75000
  )
)
customer_data
barplot(
  customer_data$Age,
  names.arg = customer_data$Customer_ID,
  col = "skyblue",
  main = "Customer Age Distribution",
  xlab = "Customer ID",
  ylab = "Age (Years)"
)
```

Output:

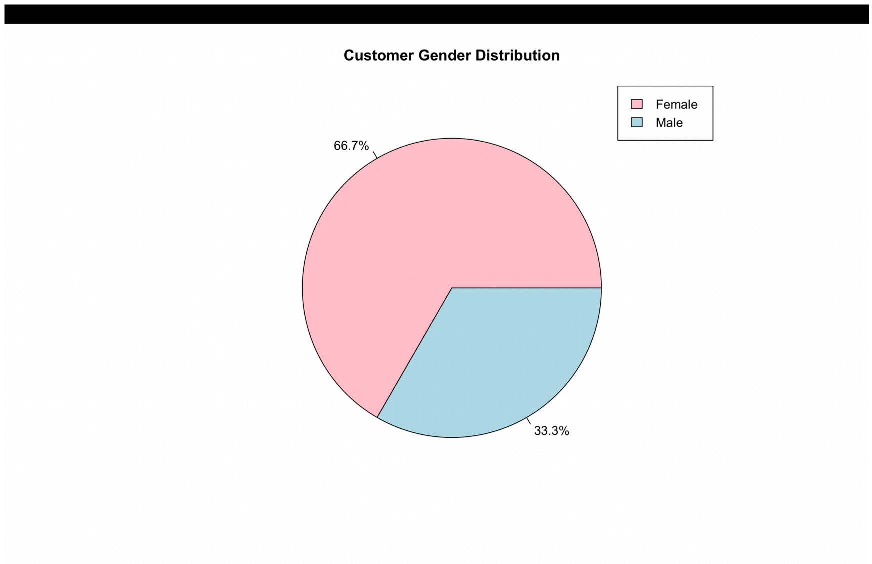


Q2. Pie Chart – Gender Distribution

Program:

```
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(
    28,
    35,
    42
  ),
  Gender = c(
    "Female",
    "Male",
    "Female"
  ),
  Income = c(
    50000,
    60000,
    75000
  )
)
customer_data
gender_count <- table(customer_data$Gender)

percentage <- round(
  gender_count / sum(gender_count) * 100,
  1
)

pie(
  gender_count,
  labels = paste0(percentage, "%"),
  col = c("pink", "lightblue"),
  main = "Customer Gender Distribution"
)

legend(
  "topright",
  legend = names(gender_count),
  fill = c("pink", "lightblue")
)
```

Output:



Q3. Build a Table to Show the Demographic Information for Each Customer

Program:

```
customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c(
    "Female",
    "Male",
    "Female"
  ),
  Income = c(
    50000,
    60000,
    75000
  )
)
customer_table <- customer_data
print(customer_table)
```

Output:

```
  Customer_ID Age Gender Income
1           1  28 Female  50000
2           2  35   Male  60000
3           3  42 Female  75000
> 
```

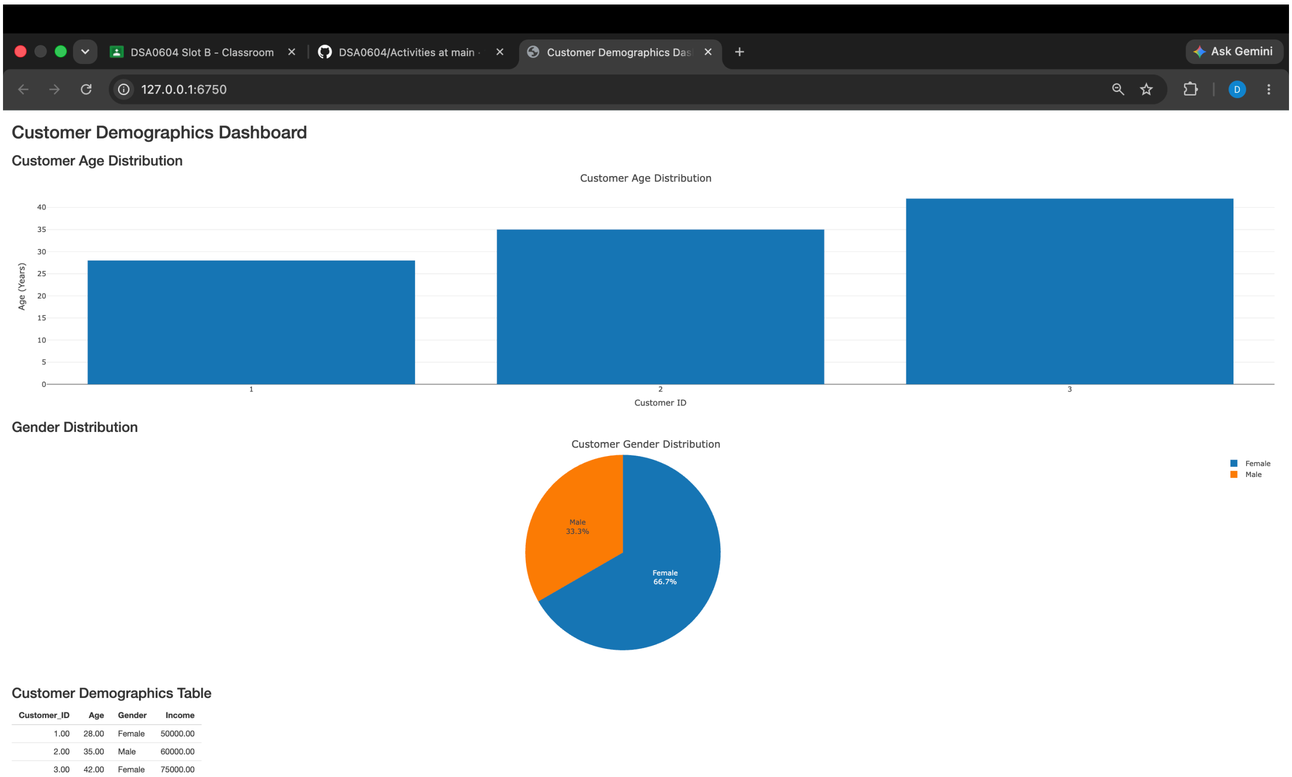
Q4. Interactive Dashboard

Program:

```
install.packages("shiny")
install.packages("plotly")

library(shiny)
library(plotly)

customer_data <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)

ui <- fluidPage(

  titlePanel("Customer Demographics Dashboard"),
  h3("Customer Age Distribution"),
  plotlyOutput("bar_chart"),
  h3("Gender Distribution"),
  plotlyOutput("pie_chart"),
  h3("Customer Demographics Table"),
  tableOutput("customer_table")
)
server <- function(input, output) {
  output$bar_chart <- renderPlotly({
    plot_ly(
      data = customer_data,
      x = ~factor(Customer_ID),
      y = ~Age,
      type = "bar"
    ) %>%
      layout(
        title = "Customer Age Distribution",
        xaxis = list(title = "Customer ID"),
        yaxis = list(title = "Age (Years)")
      )
```

```
 })
 output$pie_chart <- renderPlotly({
   gender_count <- as.data.frame(table(customer_data$Gender))
   plot_ly(
     data = gender_count,
     labels = ~Var1,
     values = ~Freq,
     type = "pie",
     textinfo = "label+percent"
   ) %>%
     layout(
       title = "Customer Gender Distribution"
     )
 })
 output$customer_table <- renderTable({
   customer_data
 })
}
shinyApp(ui = ui, server = server)
```

Output:



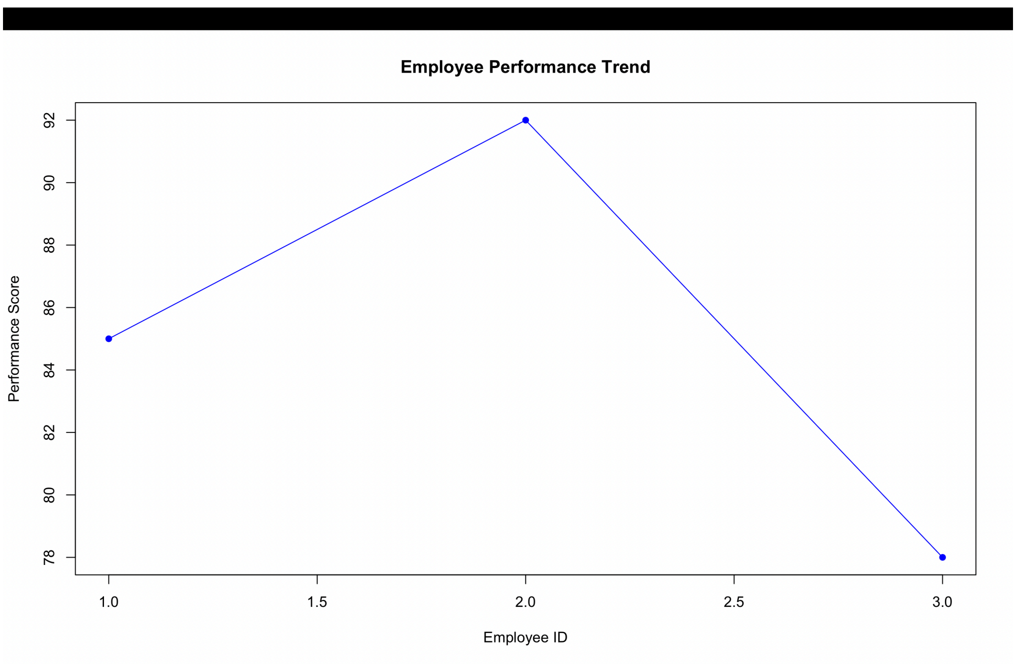
Set 17: Employee Performance Analysis

Q1. Line Chart – Employee Performance Trend

Program:

```
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
employee_data
plot(
  employee_data$Employee_ID,
  employee_data$Performance_Score,
  type = "o",
  col = "blue",
  pch = 16,
  xlab = "Employee ID",
  ylab = "Performance Score",
  main = "Employee Performance Trend"
)
```

Output:

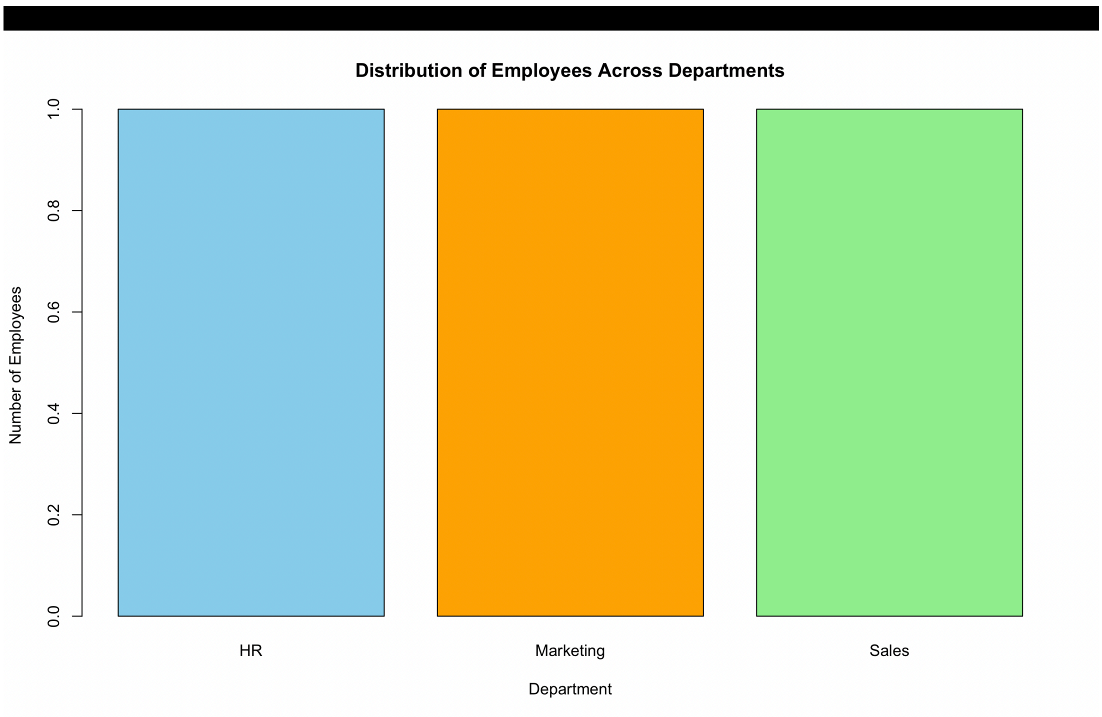


Q2. Bar Chart – Distribution of Employees Across Departments

Program:

```r
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
employee_data
department_count <- table(employee_data$Department)
barplot(
  department_count,
  col = c("skyblue", "orange", "lightgreen"),
  main = "Distribution of Employees Across Departments",
  xlab = "Department",
  ylab = "Number of Employees"
)
```

Output:



Q3. Table – Employee Performance Data

Program:

```r
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
```

```
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

employee_data
employee_table <- employee_data

print(employee_table)
```

Output:

```
  Employee_ID Department Years_of_Service Performance_Score
1           1      Sales                5                85
2           2         HR                3                92
3           3  Marketing                7                78
> |
```

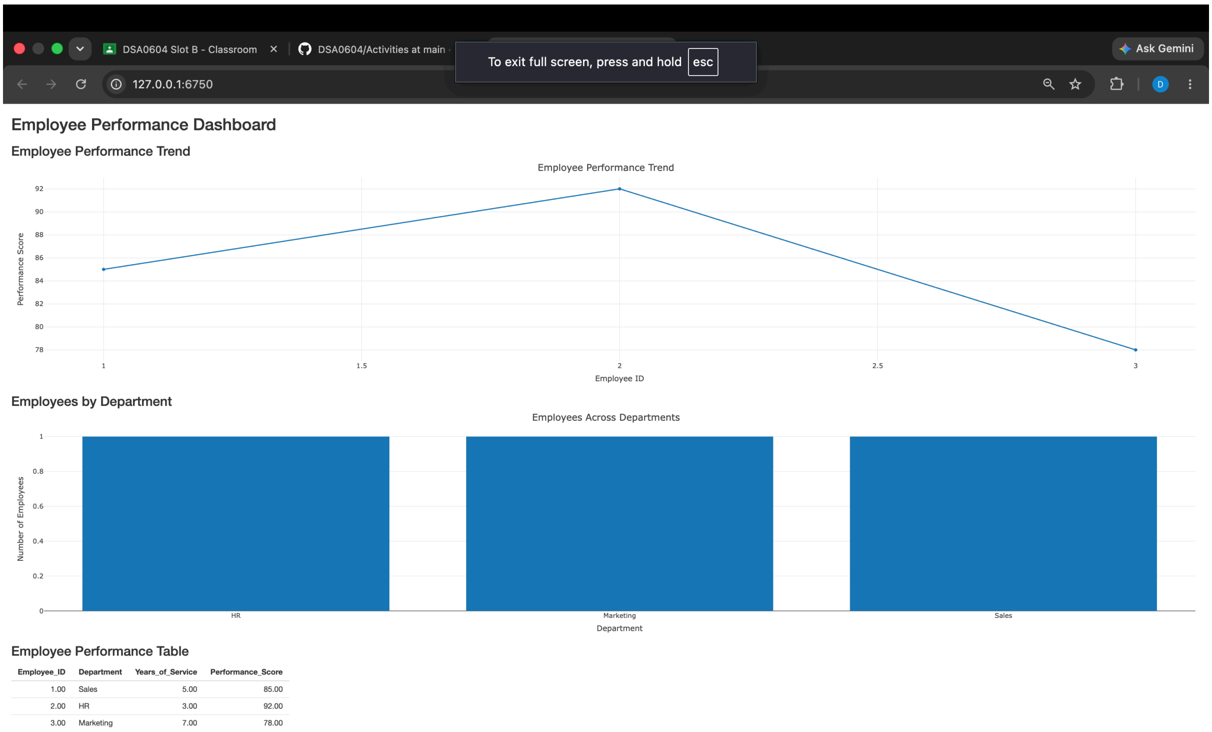
Q4. Interactive Dashboard

Program:

```
install.packages("shiny")
install.packages("plotly")

library(shiny)
library(plotly)

employee_data <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

ui <- fluidPage(

  titlePanel("Employee Performance Dashboard"),

  h3("Employee Performance Trend"),
  plotlyOutput("line_chart"),

  h3("Employees by Department"),
  plotlyOutput("bar_chart"),

  h3("Employee Performance Table"),
  tableOutput("employee_table")

)
```

```
server <- function(input, output) {

  output$line_chart <- renderPlotly({

    plot_ly(
      data = employee_data,
      x = ~Employee_ID,
      y = ~Performance_Score,
      type = "scatter",
      mode = "lines+markers"
    ) %>%

      layout(
        title = "Employee Performance Trend",
        xaxis = list(title = "Employee ID"),
        yaxis = list(title = "Performance Score")
      )

  })

  output$bar_chart <- renderPlotly({

    dept_count <- as.data.frame(table(employee_data$Department))

    plot_ly(
      data = dept_count,
      x = ~Var1,
      y = ~Freq,
      type = "bar"
    ) %>%

      layout(
        title = "Employees Across Departments",
        xaxis = list(title = "Department"),
        yaxis = list(title = "Number of Employees")
      )

  })

  output$employee_table <- renderTable({

    employee_data

  })

}

shinyApp(ui = ui, server = server)
```

Output:



Set 18: Product Inventory Management

Q1. Bar Chart – Quantity of Each Product

Program:

```
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

inventory_data
barplot(
  inventory_data$Quantity_Available,
  names.arg = inventory_data$Product_Name,
  col = c("skyblue", "orange", "lightgreen"),
  main = "Quantity of Products in Inventory",
  xlab = "Product Name",
  ylab = "Quantity Available"
)
```

Output:

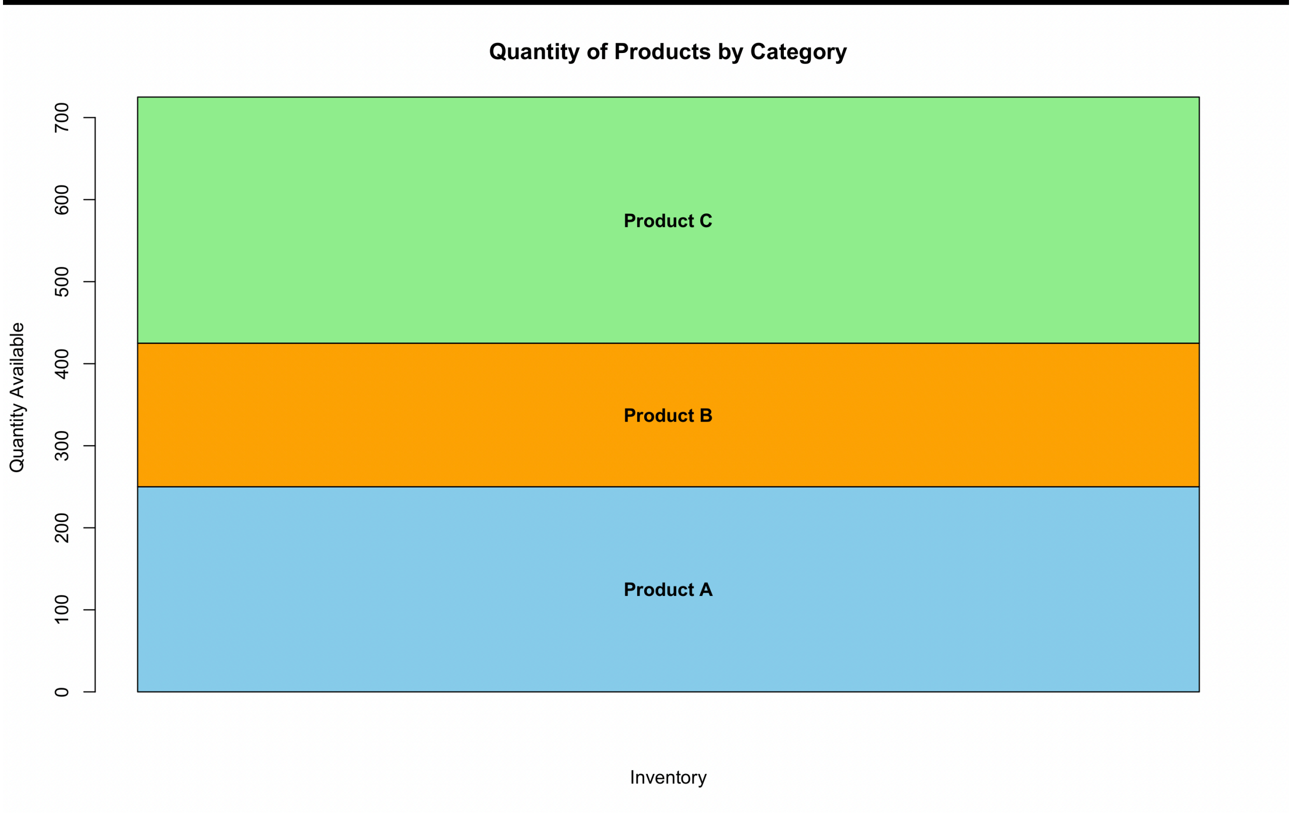
Q2. Stacked Bar Chart – Quantity by Product Category

Program:

```
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)

inventory_data
inventory_matrix <- matrix(
  inventory_data$Quantity_Available,
  nrow = 3
)

bar_pos <- barplot(
  inventory_matrix,
  beside = FALSE,
  col = c("skyblue", "orange", "lightgreen"),
  main = "Quantity of Products by Category",
  xlab = "Inventory",
  ylab = "Quantity Available"
)

text(
  x = bar_pos,
  y = c(
```

```
    250/2,
    250 + 175/2,
    250 + 175 + 300/2
  ),
  labels = c(
    "Product A",
    "Product B",
    "Product C"
  ),
  cex = 1,
  font = 2
)
```

Output:



Q3. Build a Table to Show the Inventory Data for Each Product

Program:

```
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)
```

```
inventory_table <- inventory_data

print(inventory_table)
```

Output:

```
  Product_ID Product_Name Quantity_Available Price
1          1    Product A                250    20
2          2    Product B                175    15
3          3    Product C                300    18
>
```

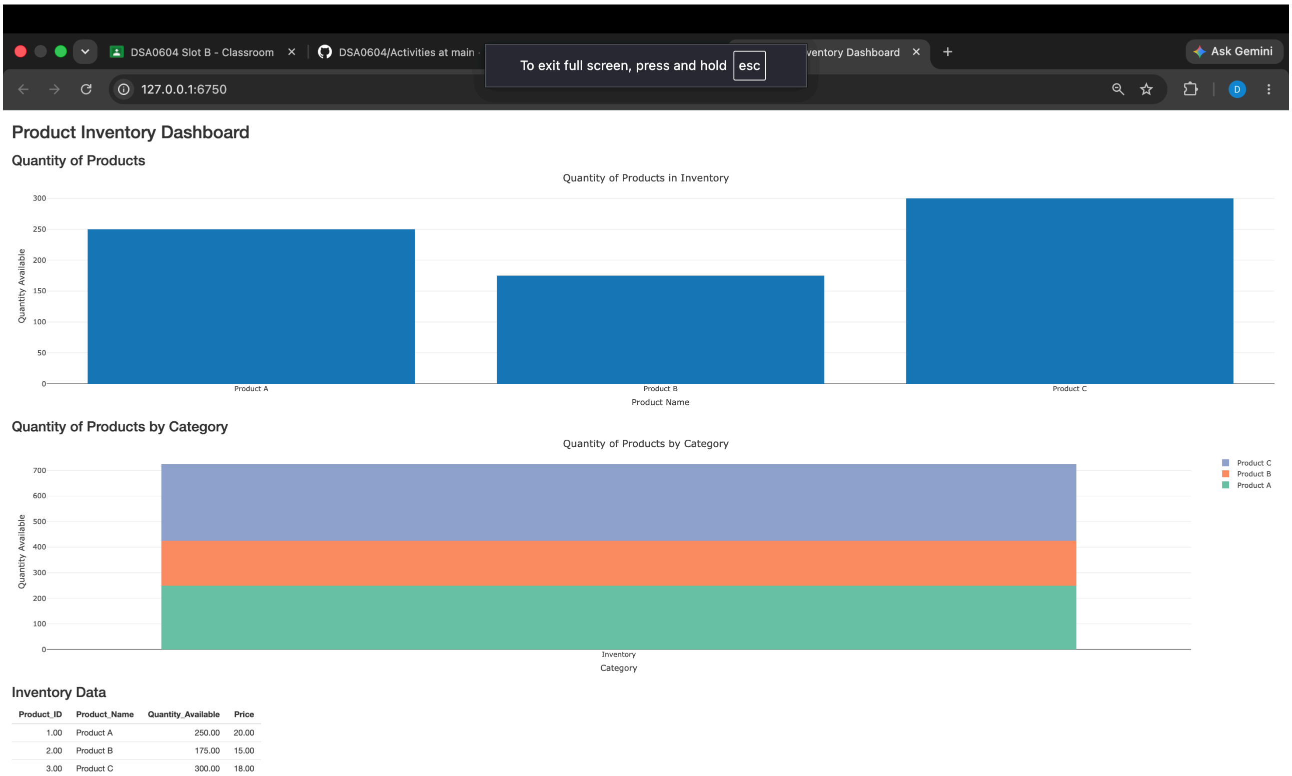
Q4. Interactive Dashboard

Program:

```
ibrary(shiny)
library(plotly)
inventory_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  Quantity_Available = c(250, 175, 300),
  Price = c(20, 15, 18)
)
ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  h3("Quantity of Products"),
  plotlyOutput("bar_chart"),
  h3("Quantity of Products by Category"),
  plotlyOutput("stacked_chart"),
  h3("Inventory Data"),
  tableOutput("inventory_table")
)
server <- function(input, output) {
  output$bar_chart <- renderPlotly({
    plot_ly(
      data = inventory_data,
      x = ~Product_Name,
      y = ~Quantity_Available,
      type = "bar"
    ) %>%
      layout(
        title = "Quantity of Products in Inventory",
        xaxis = list(title = "Product Name"),
        yaxis = list(title = "Quantity Available")
      )
  })
  output$stacked_chart <- renderPlotly({
      plot_data <- data.frame(
```

```
    Category = "Inventory",
    Product = inventory_data$Product_Name,
    Quantity = inventory_data$Quantity_Available
  )
  plot_ly(
    data = plot_data,
    x = ~Category,
    y = ~Quantity,
    color = ~Product,
    type = "bar"
  ) %>%
    layout(
      barmode = "stack",
      title = "Quantity of Products by Category",
      xaxis = list(title = "Category"),
      yaxis = list(title = "Quantity Available")
    )
 })
 output$inventory_table <- renderTable({
  inventory_data
 })
}
shinyApp(ui = ui, server = server)
```

Output:



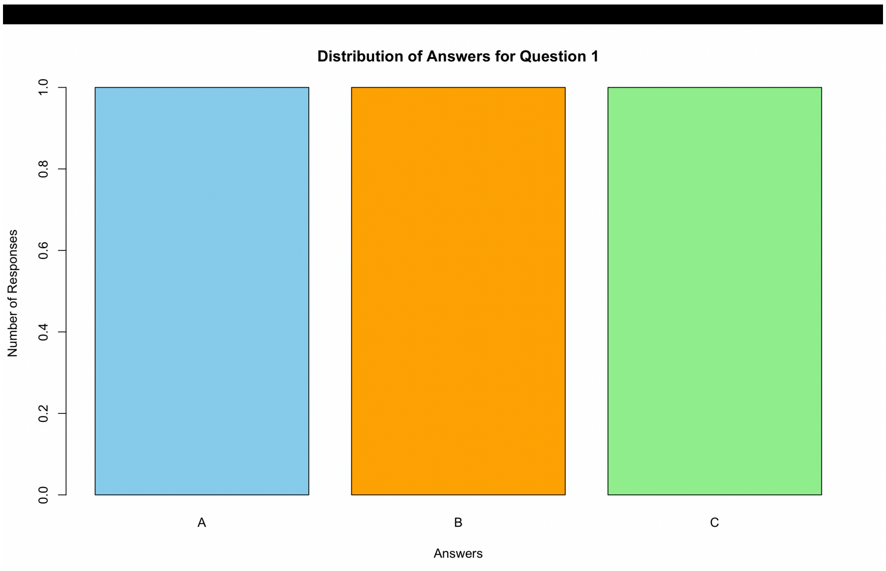
Set 19: Survey Responses Analysis

Q1. Grouped Bar Chart – Distribution of Answers for Question 1

Program:

```
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

survey_data

q1_count <- table(survey_data$Question_1)

barplot(
  q1_count,
  beside = TRUE,
  col = c("skyblue", "orange", "lightgreen"),
  main = "Distribution of Answers for Question 1",
  xlab = "Answers",
  ylab = "Number of Responses"
)
```

Output:

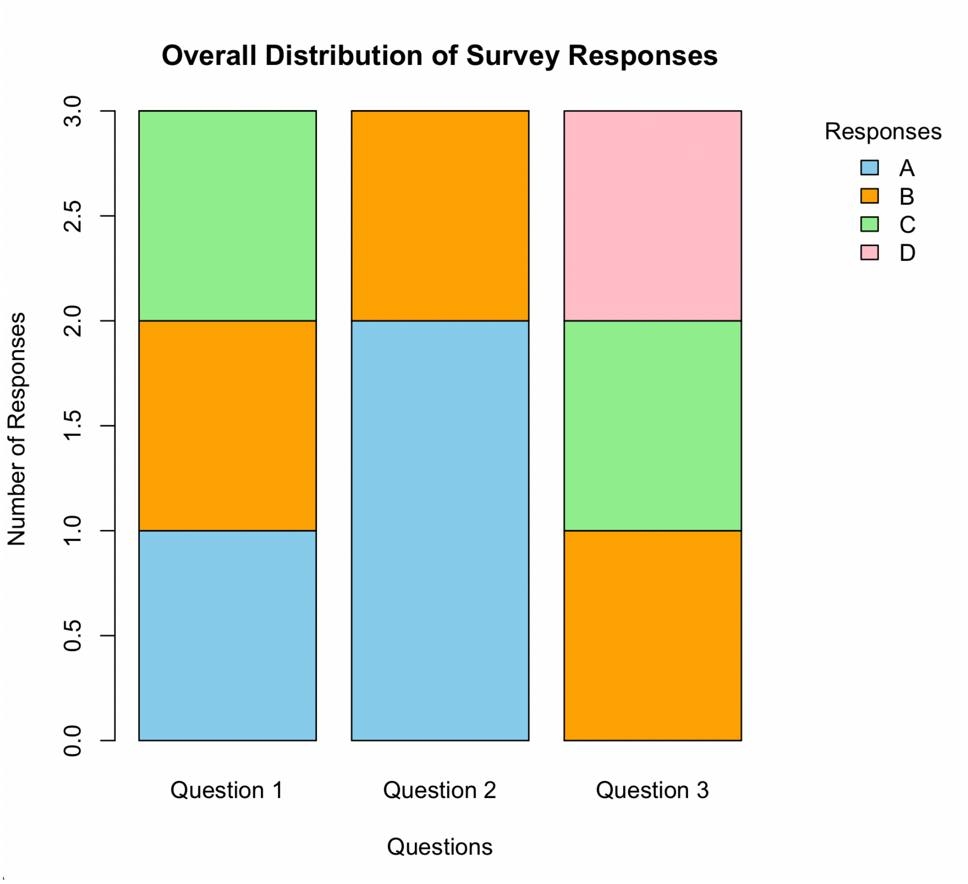


Q2. Stacked Bar Chart – Distribution of Responses for All Questions

Program:

```
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

survey_data
response_matrix <- matrix(
  c(
    1, 1, 1, 0,
    2, 1, 0, 0,
    0, 1, 1, 1
  ),
  nrow = 4,
  byrow = FALSE
)

rownames(response_matrix) <- c("A", "B", "C", "D")
colnames(response_matrix) <- c("Question 1", "Question 2", "Question 3")

par(mar = c(5, 4, 4, 8))   # Increase right margin

barplot(
  response_matrix,
  beside = FALSE,
  col = c("skyblue", "orange", "lightgreen", "pink"),
  main = "Overall Distribution of Survey Responses",
  xlab = "Questions",
  ylab = "Number of Responses"
)

legend(
  "topright",
  inset = c(-0.28, 0),
  legend = c("A", "B", "C", "D"),
  fill = c("skyblue", "orange", "lightgreen", "pink"),
  title = "Responses",
  bty = "n",
  xpd = TRUE
)
```

Output:



Q3. Build a Table to Show the Survey Response Data for Each Survey

Program:

```
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

survey_data
survey_table <- survey_data

print(survey_table)
```

Output:

```
  Survey_ID Question_1 Question_2 Question_3
1         1          A          B          C
2         2          B          A          D
3         3          C          A          B
>
```

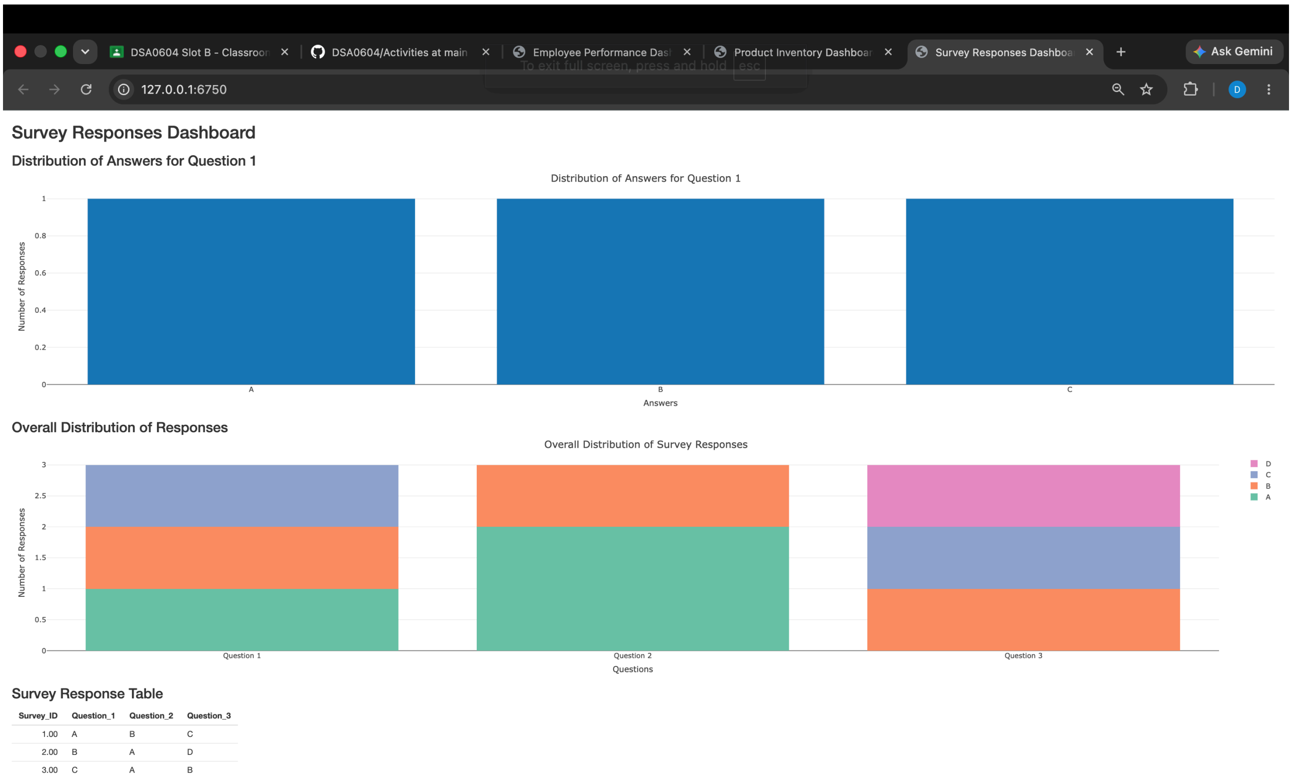
Q4. Interactive Dashboard

Program:

```
library(shiny)
library(plotly)
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)
ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  h3("Distribution of Answers for Question 1"),
  plotlyOutput("grouped_chart"),
  h3("Overall Distribution of Responses"),
  plotlyOutput("stacked_chart"),
  h3("Survey Response Table"),
  tableOutput("survey_table")
)
server <- function(input, output) {
  output$grouped_chart <- renderPlotly({
    q1 <- as.data.frame(table(survey_data$Question_1))
    plot_ly(
      data = q1,
      x = ~Var1,
      y = ~Freq,
      type = "bar"
    ) %>%
      layout(
        title = "Distribution of Answers for Question 1",
        xaxis = list(title = "Answers"),
        yaxis = list(title = "Number of Responses")
      )
  })
  output$stacked_chart <- renderPlotly({
    plot_data <- data.frame(
      Question = rep(c("Question 1", "Question 2", "Question 3"), each = 4),
      Response = rep(c("A", "B", "C", "D"), 3),
      Count = c(1,1,1,0,2,1,0,0,0,1,1,1)
    )
    plot_ly(
      data = plot_data,
      x = ~Question,
      y = ~Count,
      color = ~Response,
      type = "bar"
    ) %>%
      layout(
        barmode = "stack",
        title = "Overall Distribution of Survey Responses",
```

```
    xaxis = list(title = "Questions"),
    yaxis = list(title = "Number of Responses")
   )
 })
 output$survey_table <- renderTable({
  survey_data
 })
}
shinyApp(ui = ui, server = server)
```

Output:



Survey_ID	Question_1	Question_2	Question_3
1.00	A	B	C
2.00	B	A	D
3.00	C	A	B

Set 20: Stock Analysis

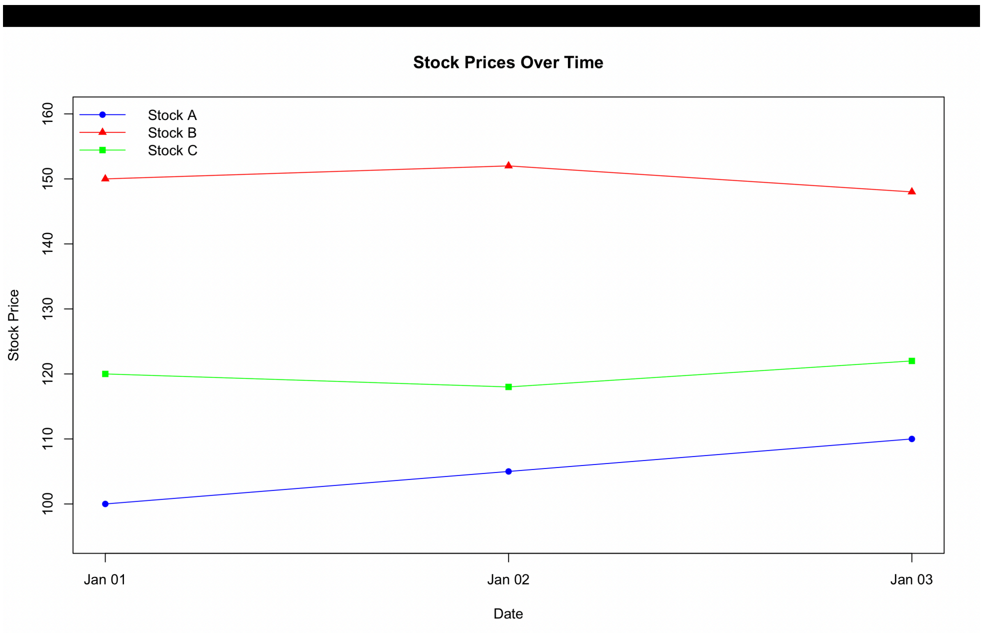
Q1. Line Chart – Stock Prices Over Time

Program:

```
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

stock_data
```

```
plot(
  stock_data$Date,
  stock_data$Stock_A,
  type = "o",
  col = "blue",
  pch = 16,
  ylim = c(95, 160),
  xlab = "Date",
  ylab = "Stock Price",
  main = "Stock Prices Over Time"
)
lines(
  stock_data$Date,
  stock_data$Stock_B,
  type = "o",
  col = "red",
  pch = 17
)
lines(
  stock_data$Date,
  stock_data$Stock_C,
  type = "o",
  col = "green",
  pch = 15
)
legend(
  "topleft",
  legend = c("Stock A", "Stock B", "Stock C"),
  col = c("blue", "red", "green"),
  lty = 1,
  pch = c(16, 17, 15),
  bty = "n"
)
```

Output:

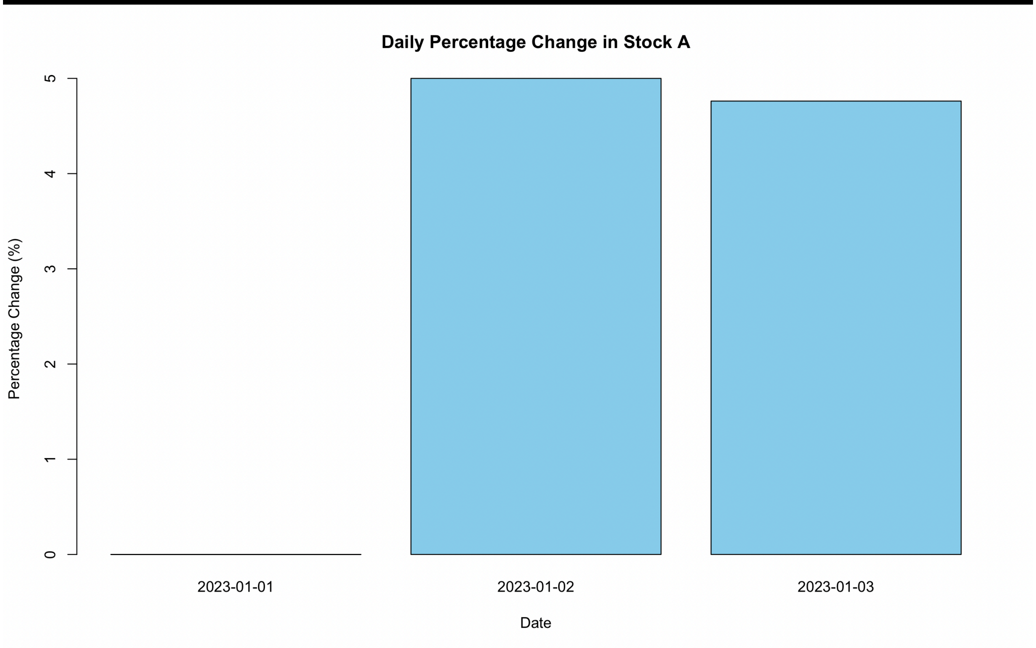


Q2. Bar Chart – Daily Percentage Change for Stock A

Program:

```
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

stock_data
percent_change <- c(
  0,
  (105 - 100) / 100 * 100,
  (110 - 105) / 105 * 100
)

barplot(
  percent_change,
  names.arg = c("2023-01-01", "2023-01-02", "2023-01-03"),
  col = "skyblue",
  main = "Daily Percentage Change in Stock A",
  xlab = "Date",
  ylab = "Percentage Change (%)"
)
```

Output:



Q3. Table – Stock Price Data

Program:

```
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

stock_data

stock_table <- stock_data

print(stock_table)
```

Output:

```
        Date Stock_A Stock_B Stock_C
1 2023-01-01     100     150     120
2 2023-01-02     105     152     118
3 2023-01-03     110     148     122
>
```

Q4. Interactive Dashboard

Program:

```
install.packages("shiny")
install.packages("plotly")

library(shiny)
library(plotly)

stock_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122)
)

ui <- fluidPage(

  titlePanel("Stock Analysis Dashboard"),

  h3("Stock Prices Over Time"),
  plotlyOutput("line_chart"),
```

```
  h3("Daily Percentage Change - Stock A"),
  plotlyOutput("bar_chart"),

  h3("Stock Price Table"),
  tableOutput("stock_table")

)

server <- function(input, output) {

  output$line_chart <- renderPlotly({

    plot_ly() %>%
      add_lines(
        x = stock_data$Date,
        y = stock_data$Stock_A,
        name = "Stock A"
      ) %>%
      add_lines(
        x = stock_data$Date,
        y = stock_data$Stock_B,
        name = "Stock B"
      ) %>%
      add_lines(
        x = stock_data$Date,
        y = stock_data$Stock_C,
        name = "Stock C"
      ) %>%
      layout(
        title = "Stock Prices Over Time",
        xaxis = list(title = "Date"),
        yaxis = list(title = "Stock Price")
      )

  })

  output$bar_chart <- renderPlotly({

    percent_change <- c(
      0,
      (105 - 100) / 100 * 100,
      (110 - 105) / 105 * 100
    )

    plot_ly(
      x = c("2023-01-01", "2023-01-02", "2023-01-03"),
      y = percent_change,
      type = "bar"
    ) %>%
      layout(
        title = "Daily Percentage Change in Stock A",
        xaxis = list(title = "Date"),
```

```
        yaxis = list(title = "Percentage Change (%)")
      )

  })

  output$stock_table <- renderTable({

    stock_data

  })

}

shinyApp(ui = ui, server = server)
```

Output: